

CURRICULUM PRIORITIZATION SURVEY

Games & VFX Art Roles

Industry Survey Analysis Report
Based on 230 Industry Professional Responses

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Executive Summary

This report synthesises findings from a curriculum prioritisation survey of **230 industry professionals** across Games (69%), Film/VFX (20%), Hybrid (6%), Education (3%), and Other (3%) sectors. Respondents are predominantly senior-level practitioners (43% Senior/Lead, 24% Director/Head), providing a hiring-manager perspective on what programmes should teach.

The survey covered 17 questions spanning respondent demographics, specialisation demand forecasts, pedagogical preferences, skills-gap analysis, and strategic curriculum advice. Below are the headline findings that emerged consistently across industries.

Headline Findings

T-Shaped graduates are the industry standard. Across every sector, the T-Shaped profile—broad pipeline awareness with one deep specialisation—was the most valued hiring outcome. Specialists ranked last.

Technical Artist is the #1 demand role in Games. When normalised for industry share, Technical Artist (Games) and Pipeline TD (Film) lead future demand. Both reflect the industry's growing need for professionals who bridge art and engineering.

Foundations before tools. A 50/50 blend of art fundamentals and digital tools in Year 1 was the clear consensus. The hybrid software-teaching model (principles first, then depth in key tools) dominated every industry segment.

AI is a tool, not a foundation. Despite heavy AI discussion in open-ended questions (~47% of emerging-tech responses), only 1.7% of respondents named AI as a non-negotiable core concept. Teamwork & Collaboration topped that list at 29%.

Industry awareness is the biggest graduate gap. At 27%, “not understanding how production actually works” was the single most cited skills deficiency, followed by technical skills (22%) and portfolio/specialisation issues (20%).

2. Recommendations

Converging signals across all 17 questions point to a clear curriculum philosophy. The following recommendations are grounded in the data. The detailed analysis supporting each recommendation follows in sections 3–13.

Foundational Architecture

- **Year 1: 50/50 Art + Digital.** Embed professional practices (version control, project management) from day one, as industry overwhelmingly prefers Year 1 introduction. Begin collaborative projects immediately with scaling complexity.
- **Principles-first software pedagogy.** Teach underlying concepts (colour theory, topology, physically-based rendering) before tool-specific workflows. Develop depth in industry-standard tools as students specialise.
- **T-Shaped graduate profile as the explicit programme outcome.** All students develop broad pipeline literacy; each student achieves demonstrable depth in one specialisation by graduation. Portfolio work should reflect this shape.

Specialisation Pathways

- **Games track:** Prioritise Technical Art, Performance & Optimisation, Real-time VFX, Rigging, Lighting, and Environment Art as specialisation options.
- **Film/VFX track:** Prioritise Pipeline TD, FX (Houdini), Compositing, Generalist Modelling, and Lighting.
- **Cross-sector disciplines:** Technical Artist, Pipeline TD, Generalist 3D Artist, Environment Art, Compositor, and Performance & Optimisation appeared across all industries and represent the safest long-term bets.

Technical Curriculum

- **Pipeline Scripting** (Python) is the #1 critical technical sub-discipline. Integrate scripting into art courses via the preferred Integrated Learning model rather than isolating it in standalone modules.
- **Houdini / procedural principles** should be offered as a specialisation pathway rather than a universal requirement (except for Film-oriented tracks where it may warrant broader coverage).

Addressing the Skills Gap

- **Industry immersion:** Tackle the #1 gap (Industry Awareness at 27%) through studio visits, guest lectures, production simulations, and industry-standard project management tools integrated from Year 1.
- **Quality benchmarking:** Establish explicit quality gates using professional reference standards. Students should regularly compare their work against shipped titles and released VFX shots.
- **Portfolio strategy:** Require focused, specialisation-aligned portfolios that demonstrate T-Shaped competence rather than unfocused generalism.

AI Integration

The data supports a **layered approach to AI**: treat it as a workflow accelerator introduced after core skills are established, not as a foundational curriculum pillar. The ~42% AI-embracing vs ~37% foundational-first split suggests a pragmatic middle path—introduce AI tools progressively alongside strengthening fundamentals, rather than making AI a standalone course or ignoring it entirely. The strongest AI advocates in the survey themselves framed it as a layer on top of existing artistry, not a standalone discipline.

DETAILED ANALYSIS

The following sections present the evidence base underpinning the recommendations above.

3. Respondent Profile

The sample provides a strong senior-practitioner perspective with good cross-industry coverage, though Games dominates. The discipline mix is well-balanced between artistic (42%) and technical (9%) orientations, with a large cohort (40%) spanning both.

Metric	Value
Total responses	230
Games	69.1%
Film / VFX	19.6%
Hybrid (both industries)	6.1%
Education	2.6%
Other (AR/VR, simulation, fine art)	2.6%
Senior / Lead	43%
Director / Head of Department	21%
Mid-level	24%
Supervisor	7%
Junior / Associate	5%

The top disciplines in **Games** were Technical Artist, Real-time VFX Artist, and Environment/Level Art. In **Film**, FX Artist (Houdini), Generalist 3D Artist, and Lighting/Look Dev Artist led. The Hybrid category was dominated by Technical Artists, confirming the cross-sector relevance of that role.

4. Specialisation Demand (3–5 Year Outlook)

When asked which specialisations will have the strongest demand for well-trained juniors, the data reveals clear per-industry priorities. Six disciplines were suggested by all industries: Technical Artist, Performance & Optimisation, Generalist 3D Artist, Pipeline TD, Environment/Level Art, and Compositor.

Games Industry (Top 6)

1. **Technical Artist** — ~28% of Games responses; dominant even after removing self-references (80% of TAs self-referenced)
2. **Performance & Optimisation** — the #2 priority, reflecting real-time rendering demands
3. **VFX Artist (Real-time)** — strong demand for Niagara/Cascade skills
4. **Rigging Artist**
5. **Lighting Artist**
6. **Environment / Level Art**

Film / VFX Industry (Top 6)

1. **Pipeline TD** — the undisputed #1 in Film; remains top even after removing self-references
2. **FX Artist (Houdini)** — close second; reflects Houdini's centrality in Film VFX
3. **Compositor** — consistently demanded across all industry segments; a cross-sector staple
4. **Generalist Modeler**
5. **Lighting Artist**
6. **Animator**

Self-Reference Bias

A consistent pattern emerged: technical/pipeline roles showed the highest self-referencing confidence (e.g., Lighting/Look Dev at 100% in Games, Rigging at 100% in Film), while artistic roles like Character Art (14%) and Concept Art (22%) rarely suggested their own discipline. Crucially, removing self-references caused only minor reordering—Technical Artist and Pipeline TD remained #1 in their respective industries—confirming these are genuine demand signals rather than self-promotion.

Notable Outliers

AI literacy appeared as a priority only from Film and Education respondents—not yet from Games. **Non-entertainment 3D** (product visualisation, fashion, footwear) was flagged as an

under-explored growth vector. Several Games respondents expressed **market saturation concerns** for traditional art roles, hinting at a tighter junior job market.

5. Critical Technical Sub-Disciplines

Pipeline Scripting was the most selected technical sub-discipline across all industries. Procedural Generation (Houdini) and Performance & Optimisation followed. Rigging & Animation Technology and Shader Authoring rounded out the main categories.

Among write-in responses, AI/ML-related mentions (AI integration, machine learning, AI error wrangling) accounted for only 3 responses—reinforcing the pattern that AI is not yet viewed as a core technical teaching priority.

6. Graduate Profile Preference

The **T-Shaped profile** was the dominant hiring preference across every industry segment. This preference was especially pronounced in Games, where the majority selected T-Shaped. Generalists ranked second overall and were most valued in Film, Hybrid, and Other categories. Specialists were the least preferred option everywhere.

This finding has direct curriculum implications: programmes should develop broad pipeline awareness in all students while providing clear pathways for deep specialisation in at least one discipline by graduation.

7. Pedagogical Preferences

Year 1 Focus

A **50/50 blend of art fundamentals and digital tools** was the clear consensus across all industries. Art Fundamentals alone came second; Technical Foundations trailed significantly. The only notable outlier was Education, where a third of respondents provided write-in answers suggesting the balance should depend on whether the student is art- or tech-focused.

Professional Practices Introduction

Most industries favour early introduction of professional practices (version control, shot tracking, asset management). Games and Film lean towards Year 1 introduction. Education was the outlier, with two-thirds preferring Year 2. This gap between educators and industry practitioners is worth noting—the people doing the hiring want these skills established earlier than the people designing curricula.

Collaborative Projects

Year 1 leads overall for starting collaborative, cross-disciplinary projects, followed closely by Year 2 and a multi-year scaling-complexity model. Again, Education diverged: no educator selected Year 1, preferring Year 2 or multi-year approaches. The Hybrid category also leaned towards Year 2.

Software Teaching Approach

The **Hybrid model** (principles first, then depth in key tools) dominated every industry. Industry Standards came second; Software-Agnostic was third. Education was the strongest advocate for the hybrid model, while Hybrid-industry respondents were the only group where no one selected Industry Standards.

Houdini / Procedural Principles

Overall, respondents favoured treating Houdini as a **Specialisation** rather than a universal requirement, driven heavily by Games. Film broke this pattern—it was the only industry where “Essential for All” led. Education respondents were the most conservative, with half viewing it as merely a Bonus Skill.

Scripting for Artists (VEX, Python)

Integrated Learning (embedding scripting within art courses) was the top preference overall. Tech Art / TD Only came second, followed by Optional Elective and Mandatory Course. Education was the strongest advocate for integration. Games showed a notably strong secondary preference for limiting scripting to Tech Art / TD students specifically.

8. Non-Negotiable Core Concepts

Analysis of 229 responses produced an 11-theme framework with a 98.3% coding rate. Because responses can carry multiple themes, percentages exceed 100%.

Theme	% of Responses
Teamwork & Collaboration	28.8%
Pipeline & Technical Understanding	18.3%
Lifelong Learning & Adaptability	16.2%
Communication & Soft Skills	15.7%
Art Fundamentals & Visual Literacy	15.3%
Production Mindset & Professionalism	10.0%
Iterative Process & Fail Fast	9.2%
Serving the Project & End Product	9.2%
Feedback Culture	5.7%
Portfolio & Career Development	4.8%
AI & Emerging Technology	1.7%

The most striking finding is that **AI & Emerging Technology ranked last at just 1.7%** (4 out of 229 responses)—despite AI featuring heavily in other survey questions. When pushed to name *the one* non-negotiable concept, professionals overwhelmingly chose human skills and artistic fundamentals. This reinforces a recurring pattern: AI is viewed as a tool layered on top of foundations, not as a foundation itself.

9. The Role of AI

The community leans cautiously optimistic. Approximately **42%** are AI-embracing (viewing AI knowledge as crucial for graduates), while **37%** are foundational-first (prioritising core artistic skills before AI exposure). Another 13.5% take a balanced view, and only ~8% are outright negative or sceptical.

The dominant tone is pragmatic (43%) or cautious (37%), with very few emotionally charged responses. Ethical and legal concerns (plagiarism, IP) were raised but only by a small minority (~1%). This suggests the industry has largely moved past the question of whether to engage with AI, and is now focused on when and how to integrate it into workflows.

10. Graduate Skills Gap

The most commonly cited gap was **Industry Awareness** (27%)—graduates not understanding how production actually works, including pipelines, deadlines, and team dynamics. This aligns with the non-negotiable concepts findings, where Teamwork & Collaboration and Pipeline Understanding together accounted for nearly half of all responses.

Skills Gap Theme	% of Responses
Industry Awareness	27.0%
Technical Skills	22.2%
Portfolio & Specialisation	20.0%
Art Fundamentals	16.5%
Quality Standards	11.7%
Soft Skills	10.0%
Production Management	8.7%
Experience Gap	8.3%
Work Ethic & Attitude	6.5%
Cross-Disciplinary Knowledge	6.5%

Two cross-cutting observations stand out. First, the specialisation crisis: respondents noted that graduates often present unfocused portfolios that try to demonstrate everything rather than excelling in one area—directly contradicting the T-Shaped preference. Second, a quality gap: graduates frequently do not recognise or meet industry quality standards, suggesting programmes need more rigorous benchmarking against professional work.

11. Emerging Technology

AI/ML dominates the emerging technology conversation at roughly 47% of responses. However, sentiment is nuanced: about 30% are explicitly positive, 14% are cautious (framing AI as a tool rather than a replacement), and ~8% are outright negative.

A strong undercurrent across responses emphasises that fundamentals and core art skills remain essential regardless of emerging tech. This mirrors the non-negotiable concepts finding—professionals see technology as something that amplifies existing skills, not something that substitutes for them.

12. Strategic Curriculum Advice

Analysis of 198 valid open-ended responses reveals eleven priority areas. The top themes reinforce the quantitative findings throughout this report.

Strategic Theme	% of Responses
Art Fundamentals & Traditional Skills	20.2%
Technical Skills & Engine Knowledge	12.1%
Curriculum Structure & Pedagogy	11.1%
Specialisation vs. Generalisation	8.6%
AI & Emerging Technology	8.1%
Real-World Preparation & Career Guidance	8.1%
Industry Collaboration & Networking	6.6%
Team Projects & Collaboration	6.6%
Portfolio Development	6.1%
Adaptability & Self-Learning	4.5%
Soft Skills & Professional Development	3.0%

The primacy of **Art Fundamentals & Traditional Skills** (20.2%) as the #1 strategic advice theme—even in an era of AI disruption—sends a clear signal. Combined with **Technical Skills & Engine Knowledge** (12.1%), these two top themes cover the hard-skill backbone that every programme must deliver.

The middle cluster (Curriculum Structure, Specialisation balance, AI, and Real-World Preparation) represents the pedagogical design challenges programmes face in structuring delivery. The lower cluster (Industry Collaboration, Team Projects, Portfolio, Adaptability, Soft Skills) maps to the enabling conditions that make hard skills industry-ready.

13. External Validation

To assess whether our survey findings reflect actual industry conditions, we cross-referenced the results against external labour market data, workforce research, and industry reports. The alignment is strong across all major findings.

Technical Artist & Pipeline TD Demand

The survey's #1 ranking for Technical Artist (Games) and Pipeline TD (Film) is corroborated by multiple external sources. Labour market analysis for 2025–2026 identifies Technical Artists, AI engineers, and gameplay programmers as the standout roles sitting at the intersection of creativity and engineering, noting that studios are looking for people who work in hybrid art-tech spaces. In Film, ScreenSkills' HETV workforce research specifically identified new shortages in technical director roles in post-production and production roles in VFX. A BFI-commissioned scoping study similarly found recruitment challenges most critical at mid-to-senior technical levels.

Market Context: The Post-2023 Reset

The broader hiring landscape adds important context to the demand data. The games industry experienced a 2% staffing contraction in 2024—the first since 2010—following roughly 10,500 layoffs in 2023 and a further 14,600 in 2025. Developer unemployment sits around 4.8%. The VFX sector faced similar pressures: after the streaming boom, studios shifted from growing headcount to optimising pipelines and budgets. Despite this contraction, specialised roles remain in demand. Global gaming employment reports note that studios still urgently need specialised talent, particularly in engine programming, technical art, and cross-platform development. The film and episodic VFX market, valued at \$5.15 billion in 2022, is projected to nearly double to \$9.91 billion by 2028, with the shortage of skilled VFX artists identified as a key market restraint.

This tighter market makes the survey's recommendations around focused portfolios, deep specialisation, and the T-Shaped graduate profile even more critical.

T-Shaped Profile & Broad Skillsets

The survey's dominant preference for T-Shaped graduates is echoed in hiring patterns. Industry reports note that candidates who can work across platforms or who have complementary cross-discipline skills are especially valuable. Job market guidance for 2026 recommends picking one technical edge—Python, Houdini, Unreal, shader work, or USD—as a differentiator, while demonstrating collaboration and pipeline awareness. This maps directly onto the T-Shaped archetype.

Skills Gap: Industry Awareness

The survey's #1 skills gap—Industry Awareness at 27%—is a widely recognised pain point. Current job market advice explicitly recommends building “production-ready” portfolios that show breakdowns, constraints, and problem-solving rather than polished beauty pieces. Academic research on the UK screen industries found persistent concerns about graduates lacking specific platform expertise and readiness for professional working conditions. ScreenSkills' regional research found insufficient industry-led training to help attain consistent standards among new entrants, and noted that crew promoted too quickly without adequate experience was fuelling skills gaps.

AI as Workflow Layer

The survey's finding that AI is viewed as a tool rather than a foundation is consistently reinforced. Industry sources frame the realistic 2026 AI impact as studios hiring people who can use AI responsibly inside a production pipeline, not those who generate content in isolation. While nearly 90% of gaming companies report using generative AI, the roles being created are hybrid positions—not standalone “AI artist” titles. IATSE and the Animation Guild's recent contract agreements include language recognising AI systems and requiring bargaining over workforce impacts, suggesting the industry is integrating AI cautiously within existing structures.

Scale of VFX Employment

The sheer scale of VFX involvement in modern production underscores the demand pipeline. Blockbusters now average over 2,000 VFX shots per film. A single feature at ILM typically involves 300–400 people working for around 18 months, outnumbering live-action crews. The global VFX workforce is estimated at approximately 50,000, while the UK screen industries alone employ over 190,000 people behind the camera. These figures illustrate both the scale of employment opportunity and the persistent need for well-trained entrants.